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H. Coprime Graph

time limit per test: 2 seconds

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an integer N . Construct an undirected graph with N nodes, where each node i is connected to all node j such that $\gcd(i, j) = 1$ where $1 \leq i, j \leq N$ and $i \neq j$.

For example, for $N = 6$, the graph will be, $G = [[2, 3, 4, 5, 6], [1, 3, 5], [1, 2, 4, 5], [1, 3, 5], [1, 2, 3, 4, 6], [1, 5]]$.

Now, there will be Q queries. Each query consists of two integers X and K . For each query, you have to determine the $K - th$ smallest node connected to node X .

Input

The first line contains two integers N and Q ($1 \leq N \leq 2 \times 10^3, 1 \leq Q \leq 3 \times 10^5$) — the number of vertices and the total number of queries.

The next Q lines contain two integers X and K ($1 \leq X \leq N, 1 \leq K \leq 10^6$), representing a query.

Output

For each query, output the $K - th$ smallest node connected to node X . If there are fewer than K neighbors of X , then print -1.

Examples

input	Copy
5 6 1 3 3 1 4 2 5 5 3 4 5 2	
output	Copy
4 1 3 -1 5 2	
input	Copy
2000 3 903 24 702 563 942 50	
output	Copy
41 1829 149	

CSE221 Assignment 04 Summer 2026

Contest is running

16:59:34

Contestant



→ Submit?

Language: Java 21 64bit

Choose file:

Choose File

No file chosen



Success!



Submit

input	Copy
1 1 1 1	
output	Copy
-1	

input	Copy
2 1 2 1	
output	Copy
1	

Note

Explanation of the First Sample (Let's go through the queries):

Query (1, 3): The neighbors of node 1 are [2, 3, 4, 5]. Sorted: [2, 3, 4, 5]. The 3rd smallest is 4.
Output: 4.

Query (3, 1): The neighbors of node 3 are [1, 2, 4, 5]. Sorted: [1, 2, 4, 5]. The 1st smallest is 1.
Output: 1.

Query (4, 2): The neighbors of node 4 are [1, 3, 5]. Sorted: [1, 3, 5]. The 2nd smallest is 3.
Output: 3.

Query (5, 5): The neighbors of node 5 are [1, 2, 3, 4]. There are only 4 neighbors, so the 5th smallest does not exist. Output: -1.

Query (3, 4): The neighbors of node 3 are [1, 2, 4, 5]. Sorted: [1, 2, 4, 5]. The 4th smallest is 5.
Output: 5.

Query (5, 2): The neighbors of node 5 are [1, 2, 3, 4]. Sorted: [1, 2, 3, 4]. The 2nd smallest is 2.
Output: 2.

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